CS2580

UNIVERSITY OF WARWICK

Second Year Winter Examinations 2012/13

Database Systems

Time allowed: 1.5 hours

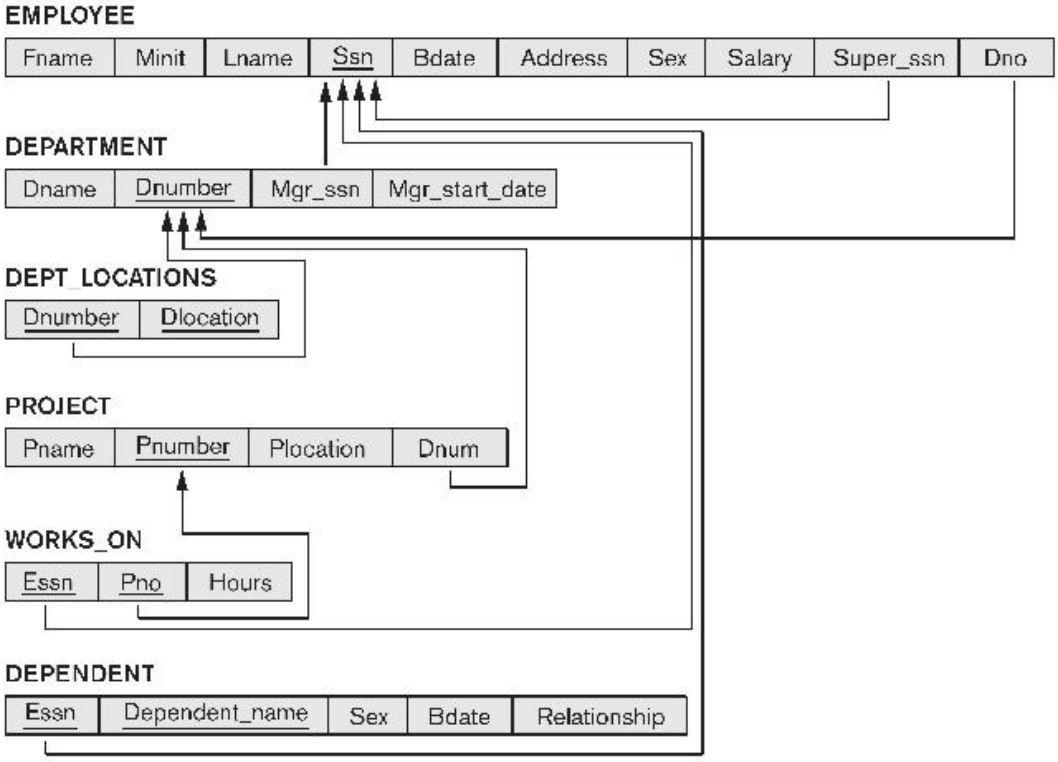
Answer **FOUR** questions.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on each answer book.

Calculators are not required and not permitted.

(continued)

1. Consider the COMPANY database whose key constraints and referential integrity constraints are depicted below.



Suppose each of the following updates is applied directly to the database state shown on the next page. Discuss all constraints violated by each operation, if any, and the different ways of enforcing these constraints.

(a) Insert < 'Robert', 'F', 'Scott', '943775543', '21-JUN-42', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1 > into EMPLOYEE. [5]
(b) Insert < 'ProductA', 4, 'Bellaire', 2 > into PROJECT. [5]
(c) Insert < 'Production', 4, '943775543', '01-OCT-88' > into DEPARTMENT. [5]
(d) Insert < '677678989', null, '40.0' > into WORKS_ON. [5]
(e) Insert < '453453453', 'John', M, '12-DEC-60', 'SPOUSE' > into DEPENDENT. [5]

(continued)

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	1965-01-09	731 Fondren, Houston, TX	M	30000	333445555	5
Franklin	T	Wong	333445555	1955-12-08	638 Voss, Houston, TX	M	40000	888665555	5
Alicia	J	Zelaya	999887777	1968-01-19	3321 Castle, Spring, TX	F	25000	987654321	4
Jennifer	S	Wallace	987654321	1941-06-20	291 Berry, Bellaire, TX	F	43000	888665555	4
Ramesh	K	Narayan	666884444	1962-09-15	975 Fire Oak, Humble, TX	M	38000	333445555	5
Joyce	A	English	453453453	1972-07-31	5631 Rice, Houston, TX	F	25000	333445555	5
Ahmad	V	Jabbar	987987987	1969-03-29	980 Dallas, Houston, TX	M	25000	987654321	4
James	E	Borg	888665555	1937-11-10	450 Stone, Houston, TX	M	55000	NULL	1

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date
Research	5	333445555	1988-05-22
Administration	4	987654321	1995-01-01
Headquarters	1	888665555	1981-06-19

DEPT_LOCATIONS

Dnumber	Dlocation
1	Houston
4	Stafford
5	Bellaire
5	Sugarland
5	Houston

WORKS_ON

Essn	Pno	Hours
123456789	1	32.5
123456789	2	7.5
666884444	3	40.0
453453453	1	20.0
453453453	2	20.0
333445555	2	10.0
333445555	3	10.0
333445555	10	10.0
333445555	20	10.0
999887777	30	30.0
999887777	10	10.0
987987987	10	35.0
987987987	30	5.0
987654321	30	20.0
987654321	20	15.0
888665555	20	NULL

PROJECT

Pname	Pnumber	Plocation	Dnum
ProductX	1	Bellaire	5
ProductY	2	Sugarland	5
ProductZ	3	Houston	5
Computerization	10	Stafford	4
Reorganization	20	Houston	1
Newbenefits	30	Stafford	4

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship
333445555	Alice	F	1986-04-05	Daughter
333445555	Theodore	M	1983-10-25	Son
333445555	Joy	F	1958-05-03	Spouse
987654321	Abner	M	1942-02-28	Spouse
123456789	Michael	M	1988-01-04	Son
123456789	Alice	F	1988-12-30	Daughter
123456789	Elizabeth	F	1967-05-05	Spouse

(continued)

2. Consider a database for hotel bookings:

   HOTEL (hotelNo, hotelName, city)
   ROOM (roomNo, hotelNo, type, price)
   BOOKING (hotelNo, guestNo, dateFrom, dateTo, roomNo)
   GUEST (guestNo, guestName, guestAddress)

   Specify the following queries in SQL:

   (a) List the numbers and names of all hotels in London. [5]
   (b) List the names and addresses of all guests living in London, alphabetically ordered by name. [5]
   (c) List the bookings for which no dateTo has been specified. [5]
   (d) List the details of all rooms at the Grosvenor Hotel, including the name of the guest staying in the room, if the room is occupied. [5]
   (e) What is the total income from bookings for the Grosvenor Hotel today? [5]

(continued)

3. Specify in SQL the following queries on the database depicted below. (Hint: you may want to use the concept of nested queries.)
   (a) Retrieve the names of all employees who work in the department that has the employee with the highest salary among all employees. [8]
   (b) Retrieve the names of all employees whose supervisor's supervisor has '888665555' for Ssn. [8]
   (c) Retrieve the names of employees who make at least $10,000 more than the employee who is paid the least in the company. [9]

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

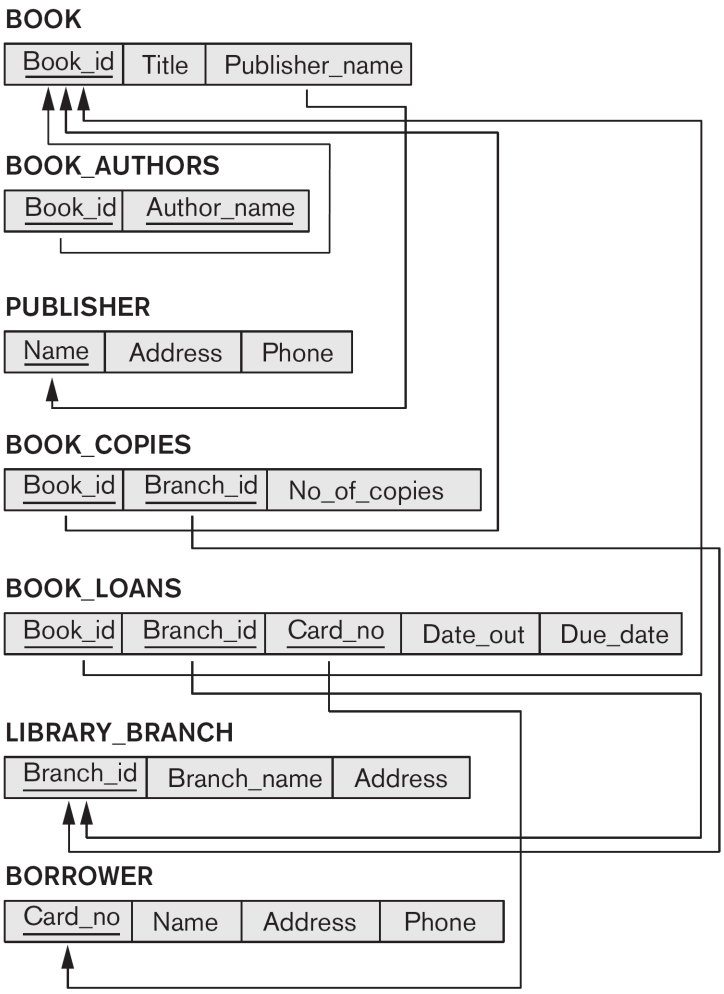
Essn	Pno	Hours

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

(continued)

4. Consider the LIBRARY relational schema shown below, where referential integrity constraints are depicted as directed arcs.



Write down expressions in relational algebra for the following queries:

(a) Retrieve the names of all borrowers who do not have any books checked out. [5]

(b) For each book loaned out from the "Sharpstown" branch and whose DueDate is today, retrieve the book title, the borrower's name, and the borrower's address. [5]

(c) For each library branch, retrieve the branch name and the total number of books loaned out from that branch. [5]

(d) Retrieve the names, addresses, and number of books checked out for all borrowers who have more than five books checked out. [5]

(e) For each book authored (or co-authored) by "Stephen King", retrieve the title and the number of copies owned by the library branch whose name is "Central". [5]

(continued)

5. An agency called *Instant Cover* supplies part-time/temporary staff to hotels within Warwickshire. The table below displays sample data, which lists the time spent by agency staff working at various hotels. The National Insurance Number (NIN) is unique for every member of staff.

   (a) The table shown is susceptible to update anomalies. Provide examples of insertion, deletion, and modification anomalies. [9]

   (b) Identify the functional dependencies represented by the attributes shown in the table. State any assumptions that you make about the data and the attributes. [6]

   (c) Normalise the table shown to 3NF. Identify primary, unique, and foreign keys in your relations. [10]

NIN	contractNo	hours	empName	hotelNo	hotelLoc
1135	C1024	16	Bird R	H25	Southam
1057	C1024	24	Foss J	H25	Southam
1068	C1025	28	Al Fayez R	H4	Leamington
1135	C1025	15	Bird R	H4	Leamington

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6. What are the main relative merits of using *discretionary access control* or *mandatory access control*? Outline also the basics of *role-based access control*. [25]

(End)